



CRANBROOK Student Number: _____
SCHOOL

2023 HSC TRIALS - ASSESSMENT TASK 4

Mathematics Extension 1

General Instructions

- Reading time – 10 minutes
- Working time – 120 minutes
- Write using blue or black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- For questions in Section II, show relevant mathematical reasoning and/or calculations

Total Marks: 70

Section I – 10 marks (pages 2–5)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 60 marks (pages 6–9)

- Attempt Questions 11–14
- Allow about 105 minutes for this section

Section I: Multiple Choice

10 marks

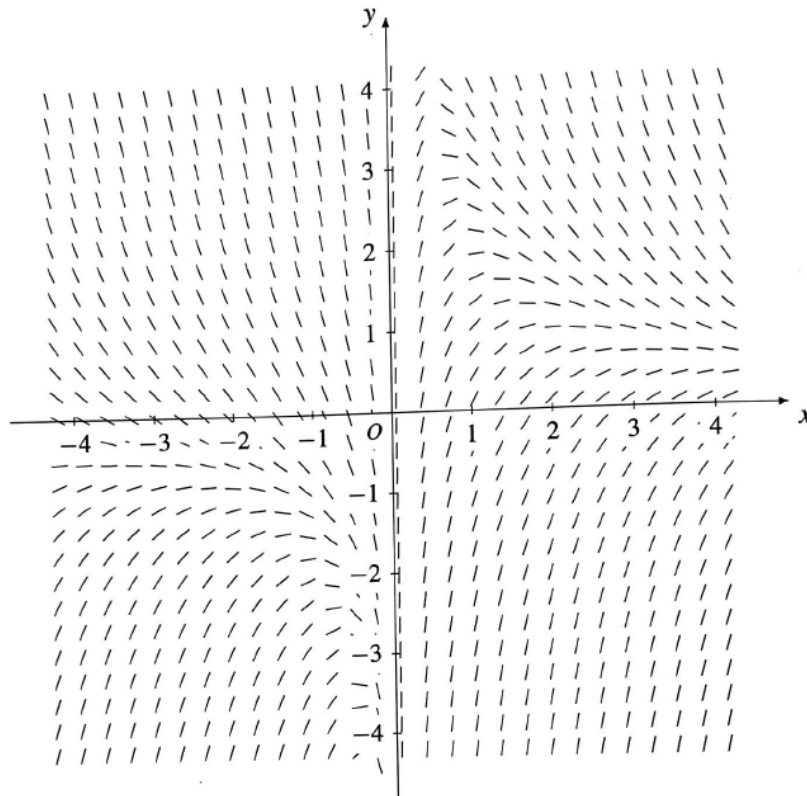
Attempt Questions 1–10

Allow about 15 minutes for this section

- 1 The polynomials $P(x)$ and $Q(x)$ are such that $P(x) = (x+3)(x-2)Q(x)$. When $P(x)$ is divided by $(x+1)$, the remainder is 12. What is the remainder when $Q(x)$ is divided by $(x+1)$?
- A. -3
B. -2
C. 2
D. 3
- 2 Which of the following is equivalent to $\frac{d}{dx} \left(2 \sin^{-1} \frac{x}{2} \right)$?
- A. $\frac{1}{\sqrt{1-x^2}}$
B. $\frac{2}{\sqrt{1-x^2}}$
C. $\frac{2}{\sqrt{4-x^2}}$
D. $\frac{1}{2\sqrt{4-x^2}}$
- 3 What is the coefficient of x^5 to $(3+2x)^7$?
- A. 21
B. 128
C. 1344
D. 6048

- 4 What is the maximum value of $15 \sin \theta - 8 \cos \theta$?
- A. 7
B. 15
C. 17
D. 23
- 5 What is the range of the function $f(x) = \tan^{-1}(\sin x)$?
- A. $\left(-\frac{\pi}{4}, \frac{\pi}{4}\right)$
B. $\left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$
C. $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
D. $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
- 6 Which of the following gives the complete domain of $y = 5 \cos^{-1}\left(\frac{2-x}{3}\right)$?
- A. $[1, 5]$
B. $[-1, 5]$
C. $[-5, 1]$
D. $[-5, -1]$
- 7 Consider the function $f(x) = x^2 + 2$, with domain $x \geq 0$.
What is the gradient of the tangent to $y = f^{-1}(x)$ at $x = 3$
- A. $\frac{1}{6}$
B. $\frac{1}{2}$
C. 2
D. 6

- 8 The slope field for a differential equation is sketched below.



Which of the following is the correct differential equation for the slope field?

- A. $\frac{dy}{dx} = \frac{2}{x} - y$
B. $\frac{dy}{dx} = \frac{2}{x} + y$
C. $\frac{dy}{dx} = \frac{2}{y} - x$
D. $\frac{dy}{dx} = \frac{2}{y} + x$

- 9** A school principal calls a meeting with two student representatives from each year group from Year 7 to Year 12. The 12 students and the principal sit at a round table. In how many ways can the 13 participants sit around the table if at least one pair of students from a year group sits apart?

- A. $12! - 6! \times 2^6$
- B. $13! - 6! \times 2^6$
- C. $12! - 5! \times 2^5$
- D. $13! - 5! \times 2^5$

- 10** Let $f(x)$ be a continuous function with $a > 0$ and $k > 0$. Which of the following is true?

- A. $\int_0^a f(x) dx = k \int_0^{ak} f(kx) dx$
- B. $\int_0^a f(x) dx = \frac{1}{k} \int_0^{ak} f(kx) dx$
- C. $\int_0^a f(x) dx = k \int_0^{ak} f\left(\frac{x}{k}\right) dx$
- D. $\int_0^a f(x) dx = \frac{1}{k} \int_0^{ak} f\left(\frac{x}{k}\right) dx$

Section II

60 marks

Attempt Questions 11–14

Allow about 1 hour 45 minutes for this section

Answer each question in this section in the separate writing booklet provided.

In Questions 11–14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (14 marks)

- (a) Solve: $|5x - 1| < 5$ **2**
- (b) $\int \frac{1}{9 + 25x^2} dx$ **2**
- (c) Find The polynomial $P(x) = x^3 + 3x^2 - 4x - 5$ has roots α, β and γ
- (i) Find $\alpha + \beta + \gamma$ **1**
- (ii) Find $\alpha^2 + \beta^2 + \gamma^2$ **2**
- (d) For the vectors $\underline{u} = 2\underline{i} + 3\underline{j}$, and $\underline{v} = 5\underline{i} - 12\underline{j}$, find each of the following:
- (i) $\underline{u} \cdot \underline{v}$ **2**
- (ii) $|\underline{v}|$ **1**
- (iii) the vector projection of $\underline{u}$ onto $\underline{v}$ **2**
- (e) In how many ways can the letters of the word POSSIBILITY be arranged so that the two Ss are together. **2**

Question 12 (15 marks)

- (a) Solve the differential equation $\frac{dy}{dx} = \frac{2}{x^3 e^y}$, where $y(1) = 0$. **3**

Express your solution in the form $y = f(x)$

- (b) Prove the identity $\tan\left(\frac{\pi}{4} + x\right) - \tan\left(\frac{\pi}{4} - x\right) = 2 \tan(2x)$ **3**

- (c) Given that $|a| = 2$, $|b| = 3$ and $a \cdot b = 5$, calculate the length of $3a - 2b$. **3**

- (d) Find the value of $\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} (\sin x - \cos x)^2 dx$ **3**

- (e) Use Mathematical induction to prove that $23^n - 1$ is divisible by 11, for all positive integers n . **3**

Question 13 (16 marks)

(a) (i) Show that $\frac{d}{dx} \tan^3 x = 3 \sec^4 x - 3 \sec^2 x$. 1

(ii) Hence, or otherwise, evaluate $\int_0^{\frac{\pi}{4}} \sec^4 x dx$ 3

(b) Find in the form $y = f(x)$ the solution of the differential equation 3

$$\left(\frac{1}{y} + \frac{1}{1-y} \right) \frac{dy}{dx} = 1$$

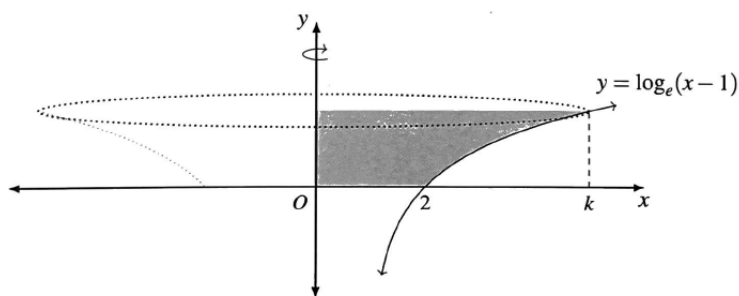
given $y = \frac{1}{2}$ when $x = 0$.

(c) (i) Show that $\cos x + \cos 5x = 2 \cos 3x \cos 2x$ 1

(ii) Hence or otherwise, solve $\cos x + \cos 5x = \cos 2x$ for $x \in [0, \pi]$. 2

(d) Use the substitution $u = \tan^{-1} x$ to find $\int_0^{\sqrt{3}} \frac{1}{(1+x^2)^{\frac{3}{2}}} dx$ 3

(e) The shaded region below is bounded by the positive x -axis and y -axis, the curve $y = \log_e(x-1)$, and a horizontal line which intersects the curve where the x -coordinate is the rational number k . 3



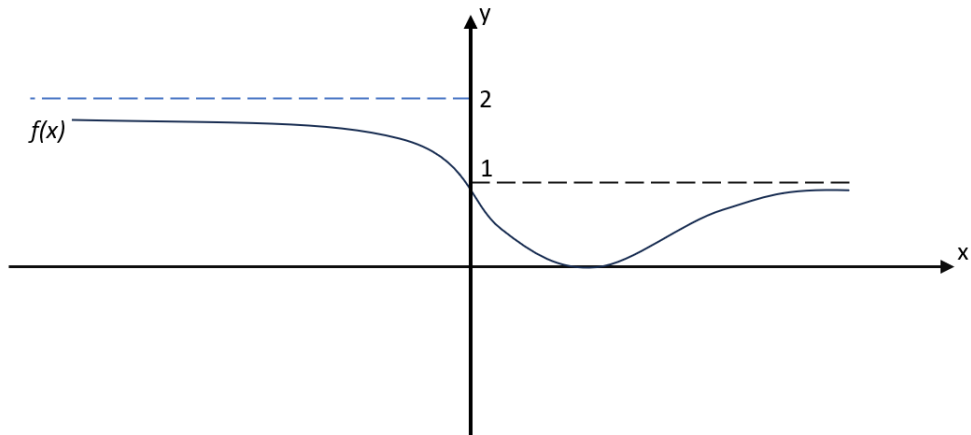
The volume V formed when this region is rotated around the y -axis is given by

$$V = \frac{\pi(27 + \log_e 16)}{2}$$

Find the value of k .

Question 14 (15 marks)

- (a) The diagram below shows the graph of $y = f(x)$.

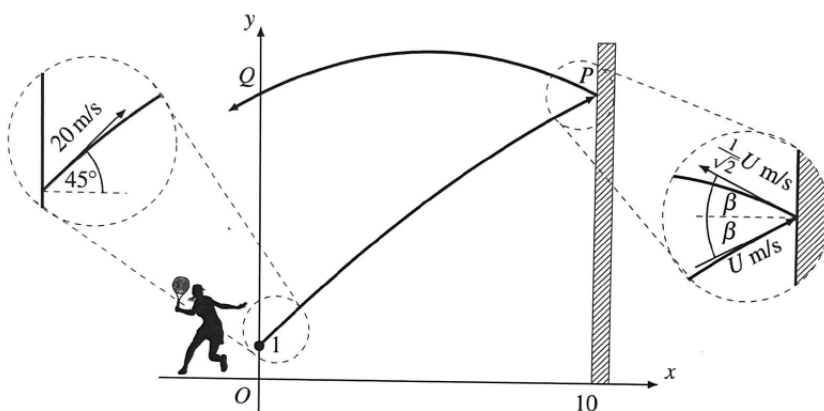


- (i) Sketch $y = \sqrt{f(x)}$ 1
- (ii) Hence, sketch $y = \frac{1}{\sqrt{f(x)}}$ on a different diagram. 2
- (b) A water tank is a prism with a base of area A square metres . Torricelli's Law states that the rate of change in the volume V cubic metres of a leaking tank with water level h metres at time t hours is given by the differential equation 5

$$\frac{dV}{dt} = -h\sqrt{h}$$

If a tank has a base with area 10 square metres and the initial height of the water is 4 metres, find how long it takes until the tank contains less than 1 litre of water.

- (c) Charlie hits a tennis ball at an angle of 45° against a vertical wall 10 m away. The ball rebounds off the wall and passes directly over where it was first struck at point Q. The tennis ball is 1 m above the ground when Charlie hits it and has an initial speed of 20 m/s.



The position $\underline{r}$ in metres of the ball t seconds after it is hit but before it hits the wall at P is given by the equation

$$\underline{r}(t) = \begin{bmatrix} 10\sqrt{2}t \\ 10\sqrt{2}t - 5t^2 + 1 \end{bmatrix}$$

- (i) Calculate the height at which the ball hits the wall. 1
- (ii) Find the velocity vector $\underline{v}(t)$ of the ball during its flight from Charlie to the wall. 1
- (iii) Calculate the speed U m/s and the angle β at which the ball hits the wall. 2
- (iv) The ball rebounds off the wall at the same angle that it hits it, β but its speed is reduced so that it leaves the wall within an initial speed of $\frac{1}{\sqrt{2}}Um/s$. 3

Show that the y - coordinates of the points P and Q in the diagram above are equal.
Assume the acceleration due to gravity is 10 m/s^2 .

End of paper

RTAY Solns

Wednesday, 9 August 2023

10:36 am

11 a) $|5x-1| < 5$

$$-5 < 5x-1 < 5 \quad (1)$$

$$-4 < 5x < 6$$

$$-\frac{4}{5} < x < \frac{6}{5} \quad (1) \checkmark$$

b) $\int \frac{1}{9+25x^2} dx$

$$= \int \frac{1}{(3)^2 + (5x)^2} dx$$

Using $\int \frac{f'(x)}{a^2 + [f(x)]^2} = \frac{1}{a} \tan^{-1}\left(\frac{f(x)}{a}\right) + C$

$$= \frac{1}{5} \int \frac{5}{(3)^2 + (5x)^2} dx \quad (1)$$

$$= \frac{1}{5} \left(\frac{1}{3} \tan^{-1}\left(\frac{5x}{3}\right) \right) + C$$

$$= \frac{1}{15} \tan^{-1}\left(\frac{5x}{3}\right) + C \quad (1) \checkmark$$

c) i) $d + \beta + \gamma = \frac{-b}{a}$

$$= \frac{-3}{1}$$

$$= -3 \quad (1) \checkmark$$

ii) $(d + \beta + \gamma)^2 = (d + \beta + \gamma)(d + \beta + \gamma)$

$$= d^2 + d\beta + d\gamma + \beta d + \beta^2 + \beta\gamma + \gamma d + \gamma\beta + \gamma^2$$

$$= d^2 + \beta^2 + \gamma^2 + 2(d\beta + \beta\gamma + d\gamma) \quad (1)$$

$$d^2 + \beta^2 + \gamma^2 = (d + \beta + \gamma)^2 - 2(d\beta + \beta\gamma + d\gamma)$$

$$= (-3)^2 - 2\left(\frac{-4}{1}\right)$$

$$= 9 + 8$$

$$= 17 \quad (1) \checkmark$$

d) i) $\underline{u} \cdot \underline{v} = x_1x_2 + y_1y_2$

$$= 2(5) + 3(-12) \quad (1)$$

$$= 10 - 36$$

$$= -26 \quad (1) \checkmark$$

MC

1) B $\checkmark$

2) C $\checkmark$

3) D $\checkmark$

4) C $\checkmark$

5) B $\checkmark$

6) B $\checkmark$

7) B $\checkmark$

8) A $\checkmark$

9) A $\checkmark$

10) D $\checkmark$

$$P(-1) = 12$$

$$(-1+3)(-1-2)Q(1) = 12$$

$$(2)(-3)Q(-1) = 12$$

$$-6Q(-1) = 12$$

$$Q(-1) = -2$$

$$\tan^{-1}(\sin x)$$

$$-1 \leq \sin x \leq 1 \quad \left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$$

$$\text{ii) } |\underline{v}| = \sqrt{5^2 + (-12)^2} \\ = 13 \quad (1) \quad \checkmark$$

$$\text{ii) Proj of } \underline{u} \text{ onto } \underline{v} \quad \frac{\underline{u} \cdot \underline{v}}{|\underline{v}|^2} \underline{v} \\ = \frac{-26}{13^2} (5\underline{i} - 12\underline{j}) \quad (1) \\ = -\frac{2}{13} (5\underline{i} - 12\underline{j}) \quad \text{OR} \quad -\frac{10}{13}\underline{i} + \frac{24}{13}\underline{j} \quad (1)$$

e) POSSIBILITY

S: 2

I: 3

Letters: 11

Bubble Method.

(11-1)! (1)

No dividing by 2! as S's aren't distinguishable.

$$\frac{10!}{3!} (1) \quad \checkmark = 604800$$

$$(12) a) \frac{dy}{dx} = \frac{2}{x^2 e^y} \\ \int e^y dy = \int 2x^{-2} dx \quad (1)$$

$$e^y = \frac{2x^{-2}}{-2} + C$$

$$e^y = -\frac{1}{x^2} + C \quad \checkmark$$

$$y(1) = 0$$

$$e^0 = -\frac{1}{1^2} + C$$

$$1 + 1 = C \quad (1)$$

$$e^y = -\frac{1}{x^2} + 2$$

$$y = \ln\left(-\frac{1}{x^2} + 2\right) \quad (1) \quad \checkmark$$

does this need absolute values?

$$b) LHS = \tan\left(\frac{\pi}{4} + x\right) - \tan\left(\frac{\pi}{4} - x\right)$$

$$= \frac{\tan\left(\frac{\pi}{4}\right) + \tan x}{1 - \tan\left(\frac{\pi}{4}\right)\tan x} - \frac{\tan\left(\frac{\pi}{4}\right) - \tan x}{1 + \tan\left(\frac{\pi}{4}\right)\tan x}$$

$$= \frac{1 + \tan x}{1 - \tan x} - \frac{1 - \tan x}{1 + \tan x}$$

$$= \frac{(1 + \tan x)^2 - (1 - \tan x)^2}{1 - \tan^2 x}$$

$$1 + 2\tan x + \tan^2 x - (1 - 2\tan x + \tan^2 x)$$

(1) expand double angle

(1) common denominator

(1) correct proof.

$$\sin^2 x + \cos^2 x = 1$$

$$= \frac{1 - \tan^2 x}{1 + 2\tan x + \tan^2 x - (1 - 2\tan x + \tan^2 x)}$$

$$\sin^2 x + \cos^2 x = 1$$

$$\tan^2 x + 1 = \sec^2 x$$

$$= \frac{4\tan x}{1 - \tan^2 x}$$

$$= \frac{4\tan x}{1 - \frac{\sin^2 x}{\cos^2 x}}$$

$$= \frac{4\sin x}{\cos x} \cdot \frac{\cos^2 x}{\cos^2 x - \sin^2 x}$$

$$= \frac{4\sin x}{\cos x} \times \frac{\cos^2 x}{\cos^2 x - \sin^2 x}$$

$$= \frac{4\sin x \cos x}{\cos^2 x - \sin^2 x}$$

$$= \frac{2(\sin 2x)}{\cos(2x)}$$

$$= \frac{2(\sin 2x)}{\cos(2x)}$$

$$= 2\tan(2x)$$

c) Distance = $|3a - 2b|$

$$\text{Distance}^2 = |3a - 2b|^2$$

$$= (3a - 2b) \cdot (3a - 2b)$$

$$= 9a^2 - 6ab + 4b^2$$

$$= 9(2)^2 - 6(5) + 4(7)^2$$

$$= 36 - 30 + 36$$

$$= 42$$

$$d = \sqrt{42}$$

$$d = 2\sqrt{3}$$

(i) Progress towards expanding length of $3a - 2b$ or equiv merit.

(i) Significant progress to finding length.

(i) Correct solution

d) $\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} (\sin x - \cos x)^2 dx$

$$= \int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \sin^2 x - 2\sin x \cos x + \cos^2 x dx$$

$$= \int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} 1 - \sin 2x dx$$

$$= \left[x + \frac{1}{2} \cos 2x \right]_{\frac{\pi}{4}}^{\frac{3\pi}{4}}$$

$$= \frac{3\pi}{4} + \frac{1}{2} \cos\left(\frac{3\pi}{2}\right) - \left(\frac{\pi}{4} + \frac{1}{2} \cos\left(\frac{\pi}{2}\right)\right)$$

$$\begin{aligned}
&= \frac{3\pi}{4} + \frac{1}{2} \omega\left(\frac{3\pi}{4}\right) - \left(\frac{\pi}{4} + \frac{1}{2} \omega\left(\frac{\pi}{4}\right)\right) \\
&= \frac{\pi}{2} + \frac{1}{2} \omega\left(\frac{3\pi}{2}\right) - \frac{1}{2} \omega\left(\frac{\pi}{2}\right) \\
&= \frac{\pi}{2} + \frac{1}{2}(0) - \frac{1}{2}(0) \\
&= \frac{\pi}{2} \quad \textcircled{1} \quad \checkmark
\end{aligned}$$

2) Prove true $n=1$

$$\text{LHS } 23^1 - 1$$

$$= 22$$

$$= 2(11)$$

$$= 11M$$

$$= \text{RHS.} \quad \textcircled{1}$$

Assume true for $n=k$

$$23^k - 1 = 11M \quad \textcircled{1}$$

Prove true for $n=k+1$

$$\text{LHS } 23^{k+1} - 1$$

$$= 23 \times 23^k - 1$$

$$= 23 \times 23^k - 1 - 22 + 22$$

$$= 23 \times 23^k - 23 + 22$$

$$= 23(23^k - 1) + 22$$

$$= 23(11M) + 22$$

$$= 11(23M) + 22$$

$$= 11(23M + 2) \quad \checkmark \quad \textcircled{1}$$

$$= 11R \quad \text{where } R \text{ is an integer}$$

As proved by MI.

$$\begin{aligned}
\textcircled{13} \text{ a) i) } \frac{d}{dx}(\tan^3 x) &= \frac{d}{dx}(\tan x) \times 3(\tan x)^2 \\
&= 3 \sec^2 x \tan^2 x \\
&= 3 \sec^2 x (\sec^2 x - 1) \\
&= 3 \sec^4 x - 3 \sec^2 x
\end{aligned}$$

$\sin^2 x + \cos^2 x = 1$
 $\tan^2 x + 1 = \sec^2 x$

keep all steps $\textcircled{1}$

$$\text{ii) } \int_0^{\frac{\pi}{4}} \sec^4 x \, dx$$

$$\frac{d}{dx}(\tan^3 x) + 3 \sec^2 x = 3 \sec^4 x$$

$$\frac{1}{3} \left[\frac{d}{dx}(\tan^3 x) \right] + \sec^2 x = \sec^4 x$$

$$\int_0^{\frac{\pi}{4}} \left(\frac{1}{3} \frac{d}{dx}(\tan^3 x) + \sec^2 x \right) dx$$

$$= \frac{1}{3} \int_0^{\frac{\pi}{4}} \frac{d}{dx}(\tan^3 x) + \int_0^{\frac{\pi}{4}} \sec^2 x$$

$$= \frac{1}{3} \left[\tan^3 x \right]_0^{\frac{\pi}{4}} + \left[\tan x \right]_0^{\frac{\pi}{4}}$$

$\textcircled{1}$ Some progress

$\textcircled{2}$ Significant progress connecting part i) to integral.

$\textcircled{3}$ Correct soln.

$$= \frac{1}{3} [1 - 0] + (1 - 0)$$

$$= \frac{1}{3} + 1$$

$$= \frac{4}{3}$$

$$b) \left(\frac{1}{y} + \frac{1}{1-y} \right) \frac{dy}{dx} = 1$$

$$\int \left(\frac{1}{y} + \frac{1}{1-y} \right) dy = \int 1 dx$$

$$\ln|y| + \int \frac{-1}{1-y} dy = x + C$$

$$\ln|y| - \ln|1-y| = x + C$$

$$\ln\left|\frac{y}{1-y}\right| = x + C$$

$$\ln\left|\frac{y}{1-y}\right| = C$$

$$C = 0$$

$$\therefore \ln|y| - \ln|1-y| = x$$

$$\ln\left|\frac{y}{1-y}\right| = x$$

$$\frac{y}{1-y} = e^x$$

$$y = e^x(1-y)$$

$$y = e^x - e^x y$$

$$y + e^x y = e^x$$

$$y(1+e^x) = e^x$$

$$y = \frac{e^x}{1+e^x}$$

$$9i) \text{ LHS} = \cos x + \cos 5x$$

$$\text{Using } \cos A \cos B = \frac{1}{2} [\cos(A-B) + \cos(A+B)]$$

$$2 \cos A \cos B = \cos(A-B) + \cos(A+B)$$

$$\therefore A+B = 5$$

$$A-B = 1$$

$$A = 3x$$

$$B = 2x$$

$$\therefore \cos x + \cos 5x$$

$$= \cos(3x-2x) + \cos(3x+2x)$$

$$= 2 \cos(3x) \cos(2x)$$

$$ii) \cos x + \cos 5x = \cos 2x$$

$$2 \cos(3x) \cos(2x) = \cos(2x)$$

$$2 \cos(3x) \cos(2x) - \cos(2x) = 0$$

$$\cos(2x) [2 \cos(3x) - 1] = 0$$

$$\cos(2x) = 0 \quad 2 \cos(3x) = 1$$

$$2x = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}$$

$$x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$$

$$3x = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}, \frac{11\pi}{3}, \frac{13\pi}{3}, \frac{17\pi}{3}$$

$$x = \frac{\pi}{9}, \frac{5\pi}{9}, \frac{7\pi}{9}, \frac{11\pi}{9}, \frac{13\pi}{9}, \frac{17\pi}{9}$$

$$\therefore x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{\pi}{9}, \frac{5\pi}{9}, \frac{7\pi}{9}, \frac{11\pi}{9}$$

$$x = \frac{\pi}{4}, \frac{3\pi}{4}$$

$$3x = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$$

$$\therefore x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$$

d) i) $\int_0^{\sqrt{3}} \frac{1}{(1+x^2)^{\frac{3}{2}}} dx$

$$u = \tan^{-1}(x)$$

$$\text{limits: } u = \tan^{-1}(\sqrt{3}) = \frac{\pi}{3}$$

$$u = \tan^{-1}(0)$$

$$= 0$$

(1) find upper/lower limits.

$$\frac{du}{dx} = \frac{1}{1+x^2}$$

$$du = \frac{1}{1+x^2} dx$$

$$= \int_0^{\frac{\pi}{3}} \frac{1}{(1+x^2)} \times \frac{1}{(1+x^2)^{\frac{1}{2}}} du$$

$$= \int_0^{\frac{\pi}{3}} \frac{1}{(1+x^2)^{\frac{3}{2}}} du$$

$$x = \tan(u)$$

$$= \int_0^{\frac{\pi}{3}} \frac{1}{(1+\tan^2 u)^{\frac{3}{2}}} du$$

$$\sin^2 x + \cos^2 x = 1$$

$$\tan^2 x + 1 = \sec^2 x$$

$$= \int_0^{\frac{\pi}{3}} \frac{1}{(\sec^2 u)^{\frac{3}{2}}} du$$

$$= \int_0^{\frac{\pi}{3}} \frac{1}{\sec^3 u} du \leftarrow (1) \text{ correct integrand}$$

$$= \int_0^{\frac{\pi}{3}} \cos^3 u du$$

$$= [\sin u]_0^{\frac{\pi}{3}}$$

$$= \sin\left(\frac{\pi}{3}\right) - \sin(0)$$

$$= \frac{\sqrt{3}}{2} \leftarrow (1) \text{ correct answer.}$$

e) $V = \pi \int x^2 dy$

$$y = \ln(x-1)$$

$$e^y = x-1$$

$$x = e^y + 1$$

$$\text{when } x=k$$

$$y = \ln(k-1)$$

$$V = \pi \int_0^{\ln(k-1)} (e^y + 1)^2 dy \quad (1)$$

$$= \pi \int_0^{\ln(k-1)} e^{2y} + 2e^y + 1 dy$$

$$= \pi \left[\frac{1}{2} e^{2y} + 2e^y + y \right]_0^{\ln(k-1)} \quad (1) \text{ correct integration OR limits}$$

$$= \frac{1}{2} e^{2\ln(k-1)} + 2e^{\ln(k-1)} + \ln(k-1) - \left(\frac{1}{2} e^{2(0)} + 2e^0 + 0 \right)$$

$$\begin{aligned}
 &= \left[\frac{1}{2} e^{2y} + 2e^y + y \right]_0 \\
 &= \frac{1}{2} e^{2h(k-1)} + 2e^{h(k-1)} + h(k-1) - \left(\frac{1}{2} e^{2e^0} + 2e^0 + 0 \right) \\
 &= \frac{1}{2} e^{h((k-1)^2)} + 2e^{h(k-1)} + h(k-1) - \frac{1}{2} - 2 \\
 &= \pi \left[\frac{1}{2} (k-1)^2 + 2(k-1) + h(k-1) - \frac{5}{2} \right]
 \end{aligned}$$

$$\frac{\pi}{2} [k^2 - 2k + 1 + 4k - 4 + 2h(k-1) - 5]$$

$$\frac{\pi}{2} [k^2 + 2k - 8 + h(k-1)^2] = \frac{\pi}{2} (27 + h(16))$$

$$k^2 + 2k - 8 = 27$$

$$k^2 + 2k - 35 = 0$$

$$(k+7)(k-5) = 0$$

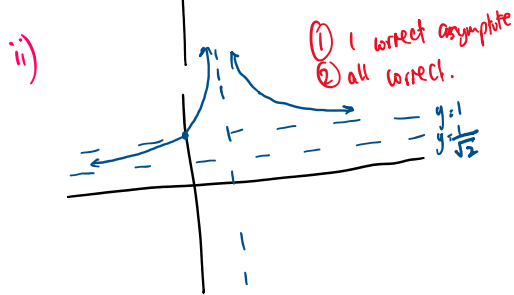
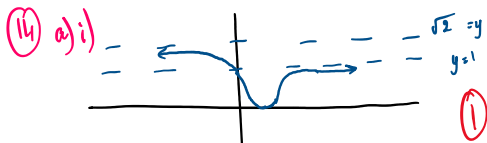
$$k = -5 \text{ or } k = 5$$

$$\text{but } h(k-1)$$

$$k-1 > 0$$

$$\therefore k = 5$$

① Correct answer.



$$b) \frac{dV}{dt} = -h\sqrt{h}$$

$$V = Ah$$

$$V = 10h$$

$$\frac{dV}{dh} = 10 \quad \text{①}$$

$$\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt} \quad \text{①}$$

$$= \frac{1}{10} \times -h^{\frac{3}{2}}$$

$$= -\frac{h^{\frac{3}{2}}}{10}$$

$$\frac{dt}{dh} = \frac{-10}{h^{\frac{3}{2}}}$$

$$\frac{dt}{dh} = -10h^{-\frac{3}{2}}$$

$$\int dt = \int -10h^{-\frac{3}{2}} dh$$

$$t = \frac{-10h^{-\frac{1}{2}}}{-\frac{1}{2}} + C \quad \text{①}$$

$$t = \frac{20}{\sqrt{h}} + C$$

$$\text{when } t=0, h=4 \quad \text{①}$$

$$\frac{20}{\sqrt{h}} + c$$

when $t=0$, $h=4$ ①

$$0 = \frac{20}{\sqrt{4}} + c$$

$$c = -10$$

$$t = \frac{20}{\sqrt{h}} - 10$$

Find when $V = 1L$

$$1L = \frac{1}{1000} m^3$$

$$V = 10h$$

$$\frac{1}{1000} = 10h$$

$$h = \frac{1}{10000}$$

$$t = \frac{20}{\sqrt{\frac{1}{10000}}} - 10$$

$$= \frac{20}{\frac{1}{100}} - 10$$

$$= 2000 - 10$$

$$= 1990 \text{ hours. } \textcircled{1}$$

d) i) $\vec{r}(t) = \begin{bmatrix} 10\sqrt{2}t \\ 10\sqrt{2}t - 5t^2 + 1 \end{bmatrix}$

$$x = 10\sqrt{2}t$$

$$10 = 10\sqrt{2}t$$

$$t = \frac{10}{10\sqrt{2}}$$

$t = \frac{1}{\sqrt{2}}$ seconds the ball hit the wall.

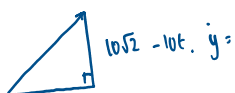
$$y = 10\sqrt{2}\left(\frac{1}{\sqrt{2}}\right) - 5\left(\frac{1}{\sqrt{2}}\right)^2 + 1$$

$$= 10 - \frac{5}{2} + 1$$

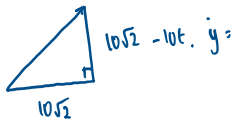
$$= 8.5m \quad \textcircled{1} \quad \checkmark$$

ii) $\vec{v}(t) = \begin{bmatrix} 10\sqrt{2} \\ 10\sqrt{2} - 10t \end{bmatrix} \quad \textcircled{1} \quad \checkmark$

iii)



iii)



at $t = \frac{1}{\sqrt{2}}$

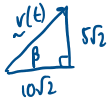
$10\sqrt{2} - 10\left(\frac{1}{\sqrt{2}}\right)$

$\dot{y} = 10\sqrt{2} - \frac{10}{\sqrt{2}}$

$\dot{y} = 10\sqrt{2} - \frac{10\sqrt{2}}{2}$

$\dot{y} = 10\sqrt{2} - 5\sqrt{2}$

$= 5\sqrt{2}$



$\tan \beta = \frac{5\sqrt{2}}{10\sqrt{2}}$

$\tan \beta = \frac{1}{2}$

$\beta = 26.565^\circ$ (1) ✓

$|\underline{U}| = \sqrt{(5\sqrt{2})^2 + (10\sqrt{2})^2}$

The $b = \sqrt{50 + 200}$

$= \sqrt{250}$

$= 5\sqrt{10}$ (1)

iv)



$v_1 = \frac{1}{\sqrt{2}} (5\sqrt{10})$

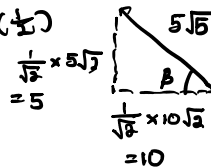
$= 5\sqrt{5}$

$\underline{a}(t) = \begin{bmatrix} 0 \\ 10 \end{bmatrix}$

$\underline{v}_1(t) = \begin{bmatrix} c_1 \\ -10t + c_2 \end{bmatrix}$

← conditions at moment of rebound — M1

$\beta = \tan^{-1}\left(\frac{1}{2}\right)$



← integrates from $\underline{a}(t)$ to find $\underline{x}(t)$

$\therefore \underline{x}_1(0) = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} -10 \\ 5 \end{bmatrix}$

given $\underline{x}(0) = \begin{bmatrix} 10 \\ 8.5 \end{bmatrix}$

Let $t=0$ be time from ball hitting wall.

at $t=0$

$c_1 = -5\sqrt{5} \cos(26^\circ)$

$c_2 = 5\sqrt{5} \sin(26^\circ)$

$\underline{v}_1(t) = \begin{bmatrix} -5\sqrt{5} \cos(26^\circ) \\ 10t + 5\sqrt{5} \sin(26^\circ) \end{bmatrix}$

$\underline{x}_1(t) = \begin{bmatrix} -5\sqrt{5} \cos(26^\circ) t + c_3 \\ 5t^2 + 5\sqrt{5} t \sin(26^\circ) + c_4 \end{bmatrix}$

at $t=0$ $c_3 = 10$ (given)

$c_4 = 8.5$ (part i)

$\underline{x}_1(t) = \begin{bmatrix} -5\sqrt{5} \cos(26^\circ) t + 10 \\ 5t^2 + 5\sqrt{5} t \sin(26^\circ) + 8.5 \end{bmatrix}$

$\underline{a}(t) = \begin{bmatrix} 0 \\ -10 \end{bmatrix}$

$\underline{v}(t) = \begin{bmatrix} c_1 \\ -10t + c_2 \end{bmatrix}$

at $t=0$

$\begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} -10 \\ 5 \end{bmatrix}$

$\underline{x}(t) = \begin{bmatrix} -10 \\ -10t + 5 \end{bmatrix}$ ← M1

integrates to get $\underline{x}(t)$ expression.

$\underline{x}(t) = \int \underline{x}(t) dt$

$$= \int \begin{bmatrix} -10 \\ -10t + 5 \end{bmatrix} dt$$

$$= \begin{bmatrix} -10t + C_3 \\ -\frac{10t^2}{2} + 5t + C_4 \end{bmatrix}$$

$$= \begin{bmatrix} -10t + C_3 \\ -5t^2 + 5t + C_4 \end{bmatrix}$$

$$\text{at } t=0$$

$$C_3 = 10$$

$$C_4 = 8.5$$

$$\hat{z}(t) = \begin{bmatrix} -10t + 10 \\ -5t^2 + 5t + 8.5 \end{bmatrix}$$

when ball hits Q

$$-10t + 10 = 0$$

$$t = 1$$

$$\hat{z}(1) = \begin{bmatrix} 0 \\ 8.5 \end{bmatrix} \text{ as required}$$

← M1
correct solution
showing same y coord.